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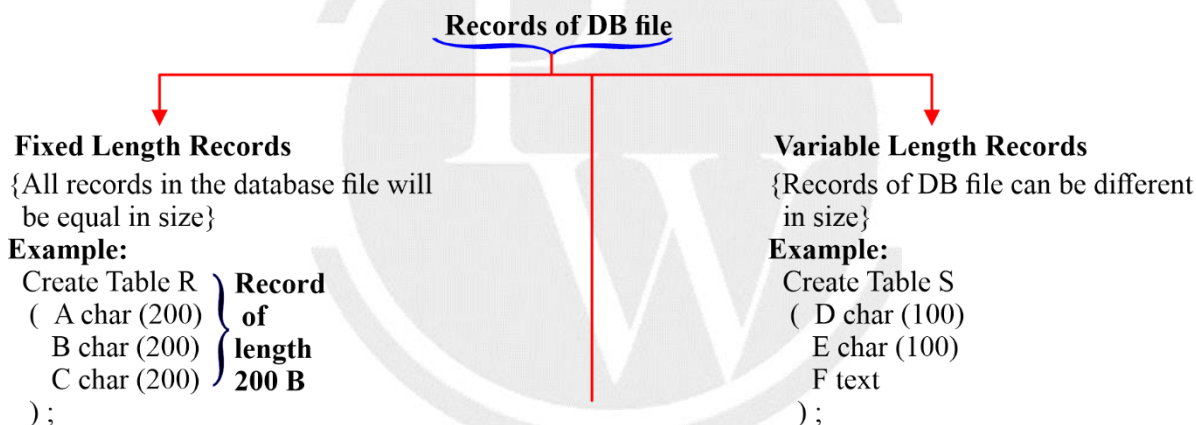
FILE ORGANIZATION AND INDEXING [B AND B⁺ TREE]

4.1 File Organization

- Database is collection of files [Tables].
- File is collection of blocks [Pages].
- Block is collection of records.

Note:

Data access from disk (DB file) to main memory block by block.



4.2 Records organization of DB file

Spanned Organization	Unspanned Organization
1. Record allowed to span in more than one block.	1. Complete record must be in one block.
2. Advantage: Possible to allocate file without any internal fragmentation.	2. Advantage: less access cost and easy to organize DB file.
3. Disadvantage: More access cost to access spanned records.	3. Disadvantage: May not possible to allocate DB file without any internal fragmentation.

Note:

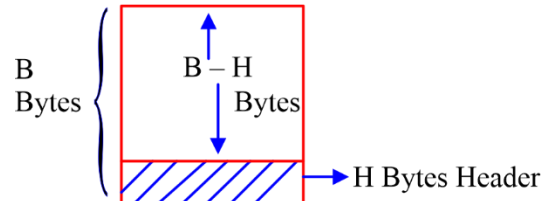
1. Spanned organization prefer to store database record of the file with variable length record.



2. Unspanned organization prefer to store DB record of file with fixed length record.

4.3 Block factor [BF]

The maximum possible records which can store in block (maximum record per block).



Consider record size = R bytes

$$1. \quad \therefore \text{Block Factor} = \left\lfloor \frac{\text{Block size} - \text{Header size}}{\text{Record size}} \right\rfloor$$

For unspanned organization

$$2. \quad \therefore \text{Block Factor} = \left\lfloor \frac{\text{Block size} - \text{Header size}}{\text{Record size}} \right\rfloor$$

For spanned organization

4.4 Categories of Index

4.4.1 Dense Index

For each DB record of DB file there must be index entry in index file.

4.4.2 Sparse Index

For set of DB records (blocks) of DB file there exists index entry in index file.

Important Point I

- Sparse index possible only over ordered field of DB file.
- In Sparse indexing

$$\left[\begin{array}{c} \text{No. of index} \\ \text{file entries} \end{array} \right] < \left[\begin{array}{c} \text{No. of records} \\ \text{of DB file} \end{array} \right]$$

4.4.3 Block Factor of Index

Assume Block size = B Bytes

Header size = H Bytes

Search key = K Bytes

Pointer size = P Bytes

$$\therefore \text{Block Factor of index} = \left\lfloor \frac{B - H}{K + P} \right\rfloor$$

Note:

By default header size will be 0 bytes, if not given in problem.

4.5 I/O Cost [Access cost]

The number of secondary memory block (DB file block) required to transfer from disk to main memory in order to access required record.

1. I/O cost to access data record without index from DB file of n blocks

- (a) Over ordered field:
Access cost = $\lceil \log_2 n \rceil$ block
- (b) Over unordered field:
Access cost = n blocks

2. If Dense Index Used:

- Number of Dense Index block = $\left\lceil \frac{\text{number of records}}{\text{Block Factor of Index}} \right\rceil$
- Access cost (Number of Block access)
= $\left\lceil \log_2 (\text{number of Dense index blocks at 1}^{\text{st}} \text{ level}) \right\rceil + 1$

3. If sparse Index Used:

- Number of DB blocks = $\left\lceil \frac{\text{number of records}}{\text{BF of DB}} \right\rceil$
- Number of sparse Index block (At 1st level)
= $\left\lceil \frac{\text{number of DB blocks}}{\text{BF of Index}} \right\rceil$
- Access cost (Number of block access)
 $\left\lceil \log_2 (\text{number of sparse index blocks at 1}^{\text{st}} \text{ level}) \right\rceil + 1$

4.6 Types of Index

- Search key: Fields of DB file which is used for indexing

